

FRYEBURG EASTERN SLOPE, ME, USA

WMO: 726183

Lat: 43.991N

Lon: 70.948W

Elev: 136

StdP: 99.71

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 54772

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.8	(c) -19.8	(d) -28.6	(e) 0.3	(f) -21.6	(g) -25.6	(h) 0.4	(i) -17.6	(j) 8.7	(k) -5.1	(l) 7.9	(m) -5.1	(n) 0.5	(o) 280	(p) 0.504

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.2	(c) 32.0	(d) 22.1	(e) 30.1	(f) 21.0	(g) 28.3	(h) 20.1	(i) 23.5	(j) 28.9	(k) 22.6	(l) 27.8	(m) 21.6	(n) 26.3	(o) 2.9	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 22.1	(b) 17.1	(c) 25.8	(d) 21.1	(e) 16.0	(f) 25.0	(g) 20.1	(h) 15.0	(i) 24.1	(j) 70.8	(k) 29.1	(l) 67.2	(m) 27.6	(n) 63.6	(o) 26.1	(p) 29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.3	(b) 6.0	(c) 5.2	(e) -29.3	(f) 34.6	(g) 3.7	(h) 1.4	(i) -31.9	(j) 35.6	(k) -34.0	(l) 36.4	(m) -36.1	(n) 37.2	(o) -38.8	(p) 38.2		
(a) 7.3	(b) 6.0	(c) 5.2	(e) -29.6	(f) 25.4	(g) 3.6	(h) 1.2	(i) -32.2	(j) 26.3	(k) -34.3	(l) 27.0	(m) -36.4	(n) 27.6	(o) -39.0	(p) 28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-7.2	-5.5	-0.7	6.3	12.8	17.6	20.6	19.6	15.5	8.6	2.5	-3.7		
	DBStd	10.76	6.60	5.73	5.64	4.54	4.19	3.81	3.10	3.09	4.20	4.42	4.91	5.64		
	HDD10.0	2190	533	434	335	131	17	1	0	0	4	80	230	424		
	HDD18.3	4249	792	668	591	363	178	59	13	21	106	301	476	683		
	CDD10.0	1192	0	0	3	18	105	228	328	299	168	38	4	0		
	CDD18.3	209	0	0	0	1	8	37	83	62	19	1	0	0		
	CDH23.3	2507	0	0	6	29	184	465	898	683	226	16	0	0		
	CDH26.7	778	0	0	1	7	57	147	294	209	62	1	0	0		
(14) Wind	WSAvg	1.7	1.7	1.9	2.2	2.2	2.0	1.7	1.4	1.3	1.3	1.6	1.7	1.6		
(15) Precipitation	PrecAvg	1220	82	87	104	100	95	117	108	98	107	112	102	111		
	PrecMax	1654	178	220	241	257	237	235	185	194	277	380	148	182		
	PrecMin	805	13	34	26	0	16	54	41	23	21	21	17	44		
	PrecStd	225	33	41	54	64	55	50	42	44	67	81	33	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	11.8	21.2	27.3	31.8	33.2	33.6	33.6	32.0	25.8	18.8	13.5		
		MCWB	9.0	6.6	12.3	15.2	19.3	22.5	23.4	22.3	22.6	16.8	13.4	9.5		
	2%	DB	7.1	7.9	14.0	22.4	28.0	30.8	31.7	31.1	28.1	22.2	15.9	9.0		
		MCWB	4.1	3.9	7.7	12.5	17.0	20.9	22.4	21.9	20.4	15.6	11.4	6.1		
	5%	DB	4.2	5.1	10.3	18.6	25.0	27.9	29.9	28.9	25.9	19.0	12.9	6.4		
		MCWB	1.3	1.9	5.0	10.5	15.4	19.3	21.2	20.8	19.2	14.1	9.7	3.5		
	10%	DB	2.1	2.9	7.6	15.7	22.4	26.1	28.0	27.3	23.6	16.6	10.5	4.0		
		MCWB	-0.1	0.1	3.5	8.5	13.9	18.1	20.4	19.9	17.7	12.5	7.4	1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	7.9	12.9	17.0	20.7	23.8	25.0	24.5	23.6	19.3	15.9	10.6		
		MCDB	10.9	10.3	18.9	23.1	28.2	30.2	31.0	29.7	29.5	22.2	17.8	12.7		
	2%	WB	4.6	4.4	8.9	14.4	18.8	22.5	23.7	23.2	21.8	16.9	12.6	6.9		
		MCDB	6.5	7.0	13.5	20.9	25.2	28.4	28.9	28.4	26.5	20.2	14.3	9.0		
	5%	WB	1.9	2.4	5.8	11.6	17.1	20.9	22.7	22.2	20.6	15.0	10.0	4.0		
		MCDB	4.0	4.8	9.3	16.2	22.5	25.8	27.7	26.8	24.0	18.3	12.9	5.9		
	10%	WB	0.1	0.7	3.8	9.2	15.4	19.6	21.7	21.2	19.1	13.2	7.8	2.2		
		MCDB	1.7	2.4	7.2	14.9	20.6	24.0	26.4	25.2	22.3	16.0	9.8	4.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.5	12.0	13.2	13.8	12.9	13.2	13.7	14.0	12.4	10.7	10.8		
		MCDBR	11.9	13.4	15.9	20.4	19.5	16.4	15.2	16.0	16.1	15.8	14.4	12.6		
	5% WB	MCWBR	9.7	10.0	10.4	11.2	9.6	7.7	7.0	7.6	8.8	9.9	10.4	10.0		
		MCWBR	11.0	12.1	14.2	17.1	15.9	14.1	13.0	12.9	13.1	13.3	12.6	11.4		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.288	0.318	0.349	0.377	0.428	0.443	0.411	0.368	0.328	0.309	0.280		
	taud		2.425	2.344	2.380	2.412	2.382	2.247	2.214	2.297	2.403	2.521	2.528	2.481		
	Ebn at Noon		871	907	921	916	894	845	826	842	856	855	817	822		
	Edh at Noon		73	94	105	111	118	135	138	123	102	80	67	64		
(44) All-Sky Solar Radiation	RadAvg		1.60	2.48	3.62	4.49	5.02	5.43	5.61	5.11	4.15	2.59	1.69	1.25		
	RadStd		0.14	0.22	0.22	0.50	0.61	0.47	0.43	0.29	0.26	0.25	0.17	0.12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+75	N/A

Nomenclature: See separate page